Write your name here

Surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-JAN18

Mathematics

Level 2



8 – 12 January 2018
Time: 1 hour 30 minutes

Paper Reference
FSM02/01

You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, compasses.

Total Marks
48/48

My signature confirms that I will not discuss the content of the test with anyone until the end of the 5 day test window.

Master Functional Skills (MFS)

Signature: _____

Instructions

- Use a **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – there may be more space than you need.
- **Calculators may be used.**

Information

- The total mark for this paper is 48.
- The marks for each question are shown in brackets
  – use this as a guide to how much time to spend on each question.
- **You must show clearly how you get your answers because marks will be awarded for your working out.**
- **Check your working and your answers at each stage.**
- **This sign shows where marks will be awarded for showing your check.**

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.

Turn over ▶

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SECTION A: Entertainment park

Answer all questions in this section.

Write your answers in the spaces provided.

1 Vinnie works at an entertainment park.

He earns £8.75 an hour.

Last week Vinnie worked for 5 days from 10 am to 6 pm.

Vinnie had one hour lunch break each day.
The hour lunch break is included in the times shown above.
Vinnie does not get paid for his lunch break.

(a) How much did Vinnie earn last week?

(3)

Use the box below to show clearly how you get your answer.

Step 1: Calculate daily working hours excluding unpaid lunch break *[M1 hours]*

Daily hours on site: 10 am to 6 pm = 8 hours total

Less 1 hour unpaid lunch break = 8 − 1 = 7 paid hours per day

Step 2: Calculate total paid working hours for the week (5 days)

Total paid hours = 5 days × 7 hours/day = 35 paid hours

Step 3: Calculate total earnings (Hours × Hourly Rate of £8.75) *[M1 earnings]*

Total Earnings = 35 hours × £8.75/hour = £306.25

Alternative calculation method check:

Gross hours = 5 × 8 = 40 hours; Gross pay = 40 × £8.75 = £350.00

Lunch deductions = 5 × £8.75 = £43.75; £350.00 − £43.75 = £306.25

Final Answer with correct money notation:

Total Earnings = £306.25 *[A1 answer]*

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Vinnie collected this information from visitors to the park

- gender (male, female)
- age (under 18, 18 – 30, over 30)
- time spent in the park (less than 2 hours, 2 – 4 hours, more than 4 hours).

Vinnie has to show this information in a data summary sheet for his manager.

(b) Design a data summary sheet for Vinnie.

(3)

Use the box below to show your data summary sheet.

Vinnie's Entertainment Park Visitor Data Summary Sheet *[M3 design]*

Time Spent	Male			Female			Total
	< 18	18–30	> 30	< 18	18–30	> 30	
< 2 hours	[tally]	[tally]	[tally]	[tally]	[tally]	[tally]	[tally]
2 – 4 hours	[tally]	[tally]	[tally]	[tally]	[tally]	[tally]	[tally]
> 4 hours	[tally]	[tally]	[tally]	[tally]	[tally]	[tally]	[tally]
Total							

Key Features of Efficient Data Collection Sheet:

- **Complete 3-way categorisation: Gender (2), Age (3), Time Spent (3).**
- **Clear input opportunities: Separate tally cell for each of the 18 sub-groups.**
- **Summary totals: Built-in total row and total column for rapid manager review.**

[A3 efficient design]

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Vinnie has to give his manager information about the number of visitors to the park.

The table below shows the number of visitors to the park each day last week.

Mon	Tue	Wed	Thu	Fri	Sat	Sun
4215	4726	4321	4952	5021	5158	4924

(c) What is the mean number of visitors to the park per day last week?
Give your answer to the nearest whole number. **(2)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total visitor count for all 7 days of the week *[M1 sum]*

Daily counts: Mon = 4215, Tue = 4726, Wed = 4321, Thu = 4952,
Fri = 5021, Sat = 5158, Sun = 4924

Total Visitors = 4215 + 4726 + 4321 + 4952
+ 5021 + 5158 + 4924 = 33,317 visitors

Step 2: Calculate mean number of visitors per day (Total ÷ 7 days) *[M1 mean]*

Mean = 33,317 ÷ 7 = 4759.571428... visitors/day

Step 3: Round answer to the nearest whole number

First decimal place is 5 (0.5714... ≥ 0.5), so round UP to next whole number.

4759.5714... □ 4760 visitors

Reverse multiplication check:

4760 × 7 = 33,320 visitors (approx equal to 33,317 total sum, diff = 3)

Final Answer (nearest whole number):

Mean visitors per day = 4760 *[A1 answer]*

(Total for Question 1 is 8 marks)

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2 Vinnie updates the website for the park.
He has to estimate the area a new ride will need.

The diagram shows the rectangular space the ride needs.

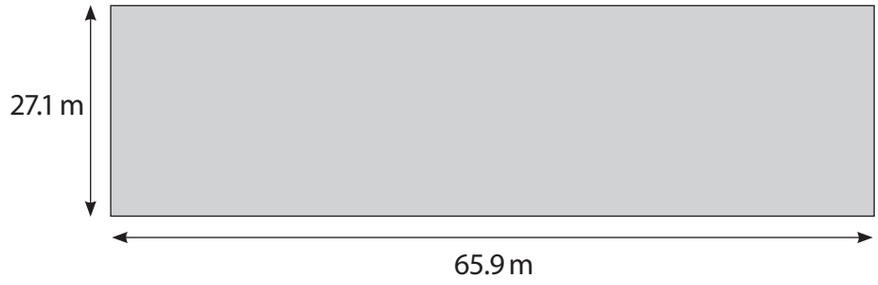


Diagram **not** accurately drawn

(a) Estimate the area of the space the ride needs.

(2)

Use the box below to show clearly how you get your answer.

Step 1: Round rectangular dimensions to suitable values for estimation *[M1 round]*

Length: 65.9 m ≈ 66 m (or to 1 s.f. = 70 m)

Width: 27.1 m ≈ 27 m (or to 1 s.f. = 30 m)

Step 2: Calculate estimated area with correct units (m^2) *[A1 units]*

Estimated Area = 66 m × 27 m = 1,782 m^2

(Alternative 1 s.f. estimation: 70 m × 30 m = 2,100 m^2)

Final Answer: 1,782 m^2 (or 2,100 m^2)

(b) Explain if your estimation is an underestimation or an overestimation.
You must give a reason for your answer.

(1)

Use the box below to write your answer.

Evaluation of Estimation (Over or Under): *[B1 reason]*

Using 1,782 m^2: Underestimation. 65.9 was rounded up by 0.1 but 27.1 was rounded down by 0.1, giving 1,782 m^2 < actual area of 1,785.89 m^2.

(If using 2,100 m^2: Overestimation because both 65.9 and 27.1 were rounded up).

(Total for Question 2 is 3 marks)

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3 There is a special offer for tickets to the entertainment park.

	Adult (ages 18 and over)	Junior (ages 12 – 17)	Child (ages 4 – 11)
1-day ticket	£55	£42	£26
2-day ticket	£90	£70	£45
3-day ticket	£130	£95	£62

Buy your tickets in January and get 37% off the total price.

In the Smith family there are 2 adults, 2 juniors and 1 child.
They want to buy a 3-day ticket for everyone in the family.
They buy the tickets in January.

The Smiths also want to book a family room at a hotel for 2 nights.
The family room costs £143.97 per night.

What is the total cost of the tickets and the family room for the Smiths?

(5)

Use the box below to show clearly how you get your answer.

Step 1: Calculate total 3-day ticket price before discount for the 5 family members *[M1 tickets]*

- 2 Adults (ages 18+): 2 × £130 = £260.00
- 2 Juniors (ages 12–17): 2 × £95 = £190.00
- 1 Child (ages 4–11): 1 × £62 = £62.00

Total Full Ticket Cost = £260 + £190 + £62 = £512.00

Step 2: Apply 37% January discount to the total ticket price *[M1 discount]*

Discount = 37% of £512.00 = 0.37 × £512 = £189.44

Discounted Tickets = £512.00 − £189.44 = £322.56

(Alternative: 0.63 × £512 = £322.56)

Step 3: Calculate hotel accommodation cost (2 nights @ £143.97/night) *[M1 hotel]*

Hotel Room Cost = 2 nights × £143.97 = £287.94

Step 4: Total Cost = £322.56 + £287.94 = £610.50 *[A1 answer]*

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Q3 Continuation: Alternative Itemised Discount Method & Verification Check

Alternative Method: Discount each ticket category first (paying 63% = 100% – 37

- 2 Adults discounted: 2 × (£130 × 0.63) = 2 × £81.90 = £163.80
- 2 Juniors discounted: 2 × (£95 × 0.63) = 2 × £59.85 = £119.70
- 1 Child discounted: 1 × (£62 × 0.63) = 1 × £39.06 = £39.06

Total Discounted Tickets = £163.80 + £119.70 + £39.06 = £322.56 (matc

Combined Family Trip Expense Summary:

- 3-day theme park tickets (5 people with 37% off): £322.56
- 2 nights family room accommodation: £287.94

Grand Total Payable = £322.56 + £287.94 = £610.50

Final Answer:

Total Cost for the Smiths = £610.50 [A1 final]

(Total for Question 3 is 5 marks)

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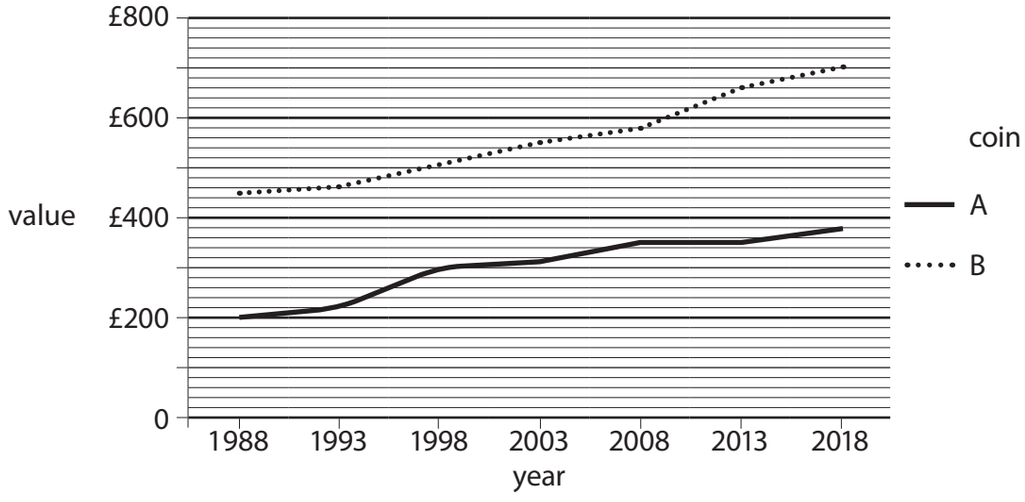
SECTION B: Collecting coins

Answer all questions in this section.

Write your answers in the spaces provided.

4 Tina collects rare coins.

This graph shows the value, over time, of two coins that Tina likes.



Tina wants to buy the coin with the highest percentage increase in value from 1988 to 2018

She uses this formula to find the percentage increase in the value of coin A.

$$V = \frac{H - L}{L} \times 100$$

V is the percentage increase in value
H is the most recent value (£)
L is the value in 1988 (£)

Tina knows that coin B increased in value by 55.5%

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(a) Which coin has the highest percentage increase in value?

(4)

Use the box below to show clearly how you get your answer.

Step 1: Read values of Coin A from the line graph *[M1 readings]*

- Initial value in 1988 (L) = £200
- Most recent value in 2018 (H) = £380

Absolute increase in value = £380 − £200 = £180

Step 2: Calculate percentage increase (V) using given formula V = [(H − L) ÷ L] × 100 *[M1 formula]*

V = [(380 − 200) ÷ 200] × 100 = (180 ÷ 200) × 100

V = 0.90 × 100 = 90% increase for Coin A

Step 3: Compare percentage increase of Coin A with Coin B (55.5%)

- Coin A percentage increase = 90%
- Coin B percentage increase = 55.5%

90% > 55.5% (Coin A increased by 34.5% more than Coin B)

Coin B verification check from graph:

Coin B: 1988 = £450, 2018 = £700; Increase = £250; (250 ÷ 450) × 100 = 55.5%

Conclusion & Final Answer:

Conclusion: Coin A has the highest percentage increase (90% > 55.5%). *[A1 decision]*

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Tina wants to buy a different coin.
The price of the coin is £850

There are two payment options.

option A	option B
pay $\frac{1}{5}$ of the price as a deposit and then 12 instalments of £65.49	pay nothing for a year then pay full price and 13% interest

Tina wants to use the cheapest option.

(b) Which option should Tina use?
Show why you think this. **(5)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total cost using Option A (1/5 deposit + 12 instalments of £65.49) [M1 opt A]

- Deposit = 1/5 of £850 = £850 ÷ 5 = £170.00
- 12 monthly instalments = 12 × £65.49 = £785.88

Total Cost Option A = £170.00 + £785.88 = £955.88

Step 2: Calculate total cost using Option B (Full price £850 + 13% interest) [M1 opt B]

- 13% Interest = 13% of £850 = 0.13 × £850 = £110.50

Total Cost Option B = £850.00 + £110.50 = £960.50 (or 1.13 × £850

Step 3: Compare costs (Tina wants the cheapest option) [M1 compare]

£955.88 < £960.50 (Option A is cheaper by £4.62)

(Difference: £960.50 − £955.88 = £4.62)

Final Decision & Recommendation:

Conclusion: Tina should use Option A. [A1 decision]

Option A (£955.88) is cheaper than Option B (£960.50) by £4.62.

(Total for Question 4 is 9 marks)

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5 Tina wants to design a closing silver box for her coin.

She wants the box to be in the shape of a cuboid with

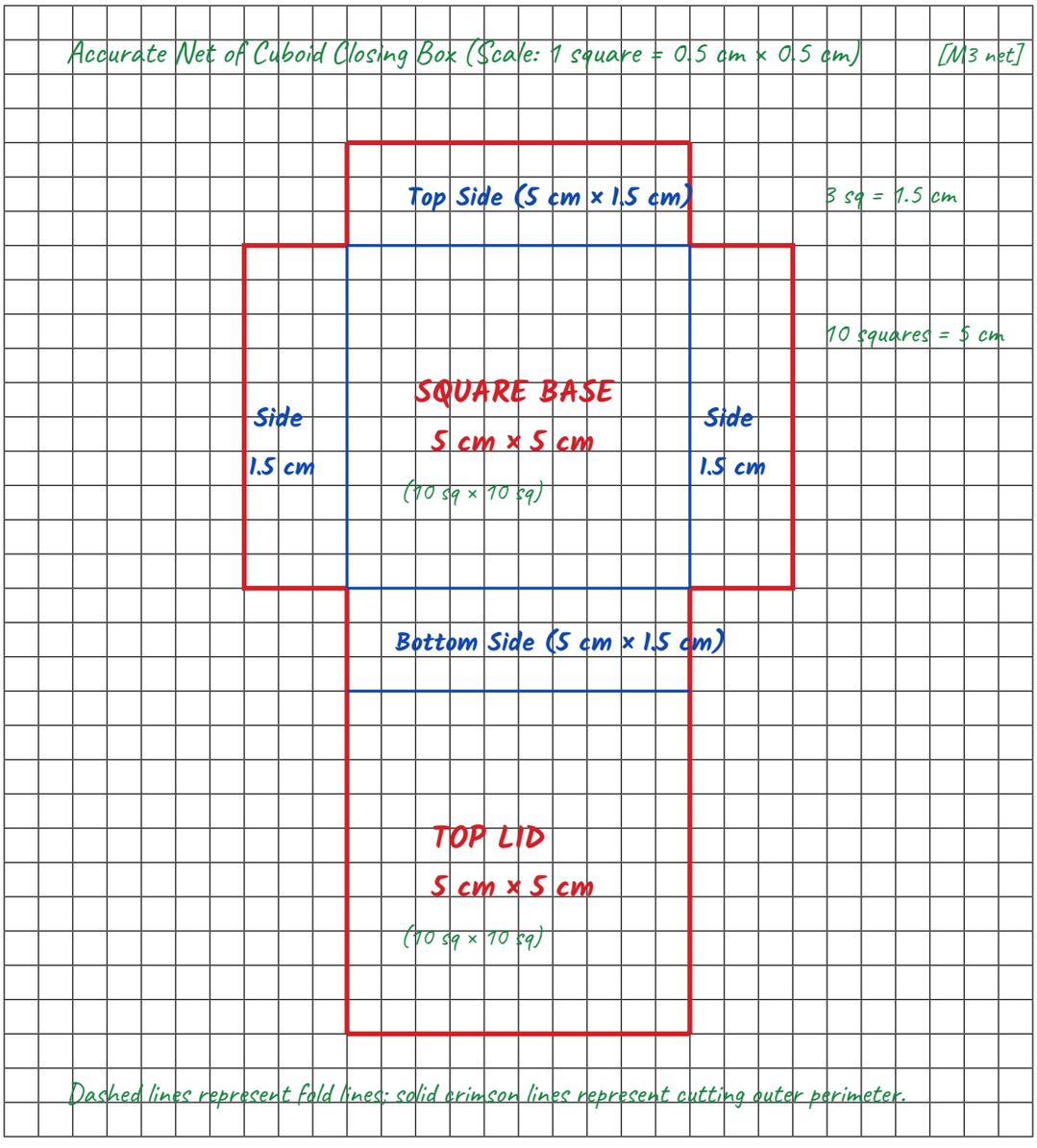
- square base of side length 5 cm
- height of 1.5 cm.

Tina needs to draw an accurate net of the box for a jeweller.

Draw an accurate net of the box for Tina.

(3)

Use the grid below to draw the net.



(Total for Question 5 is 3 marks)

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6 Tina wants to join a coin collectors club.

The table shows the club membership options

Membership type	1 month	3 months	6 months	12 months
Silver	$19	$55	$102	$199
Gold	$39	$109	$199	$359
Platinum	$59	$149	$279	$499

Tina wants to buy Gold membership for 6 months.

She knows £1 = $1.286

Tina thinks this membership will cost less than £150

Is Tina correct?
Show why you think this.
Show a check of your working **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Identify Gold 6-month membership price from the table *[M1 price]*

Table reading: Gold Membership for 6 months = $199 USD

Step 2: Convert $199 USD to British Pounds (£1 = $1.286 USD) *[M1 convert]*

Cost in £ = $199 ÷ 1.286 = £154.74339... ≈ £154.74

Step 3: Compare calculated cost with Tina's £150 estimate

£154.74 > £150.00 (membership costs £4.74 MORE than £150)

Alternative USD conversion method check:

Convert £150 to USD: £150 × 1.286 = $192.90 USD.

Actual price $199 USD > $192.90 USD ($199 > $192.90).

Conclusion & Decision:

Conclusion: No, Tina is NOT correct. *[A1 decision]*

£154.74 is greater than £150 (£154.74 > £150.00).

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Use the box below to show your check.

☑

Reverse calculation check: *[B1 check]*

£154.74 × 1.286 = $198.995... ≈ $199 USD (matches official table price)

Alternative check: $192.90 ÷ 1.286 = £150.00 (verified)

(Total for Question 6 is 4 marks)

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SECTION C: Meeting venue

Answer all questions in this section.

Write your answers in the spaces provided.

7 Cora is the manager at a meeting venue.
She organises a conference for a local business.

She has these notes.

Conference lasts 2 days

87 people attend each day

Things needed

• 3 large rooms each day	£459 per room per day
• catering	£35 per person per day
• IT support	£94.99 per conference

Cora thinks the total price for the conference will be less than £9000

(a) Is Cora correct?
Show why you think this. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate room hire cost for 2 days (3 large rooms each day @ £459/day/room) [M1 rooms]

Room Hire = 3 rooms × £459/day × 2 days = £2,754.00

Step 2: Calculate catering cost for 2 days (87 people @ £35/person/day) [M1 catering]

Catering = 87 people × £35/day × 2 days = £6,090.00

Step 3: Calculate total conference cost including IT support (£94.99 one-off) [M1 total]

Total Price = £2,754.00 + £6,090.00 + £94.99 = £8,938.99

Step 4: Compare total price with Cora's £9000 budget

£8,938.99 < £9,000.00 (under budget by £61.01)

Conclusion: Yes, Cora is correct (Total £8938.99 < £9000). [A1 decision]

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Q7(a) Continuation: Daily Breakdown & Alternative Verification Method

Daily cost calculation method (Daily budget = £9000 ÷ 2 = £4500):

- Daily Room Hire (3 rooms): 3 × £459 = £1,377.00
- Daily Catering (87 people): 87 × £35 = £3,045.00
- Daily IT support allocation: £94.99 ÷ 2 = £47.495

Daily Cost = £1,377.00 + £3,045.00 + £47.495 = £4,469.495

Two-Day Total = £4,469.495 × 2 = £8,938.99 (verified)

Budget Surplus Summary:

- Target Maximum Budget = £9,000.00
- Total Calculated Invoice = £8,938.99
- Remaining Unspent Funds = £61.01

Final Answer: Yes, Cora is correct (Total cost is £8,938.99 < £9,00

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Cora also organises a fashion show at the venue.
She must work out if the room is big enough for the fashion show.

This is a diagram of the floor in the room.

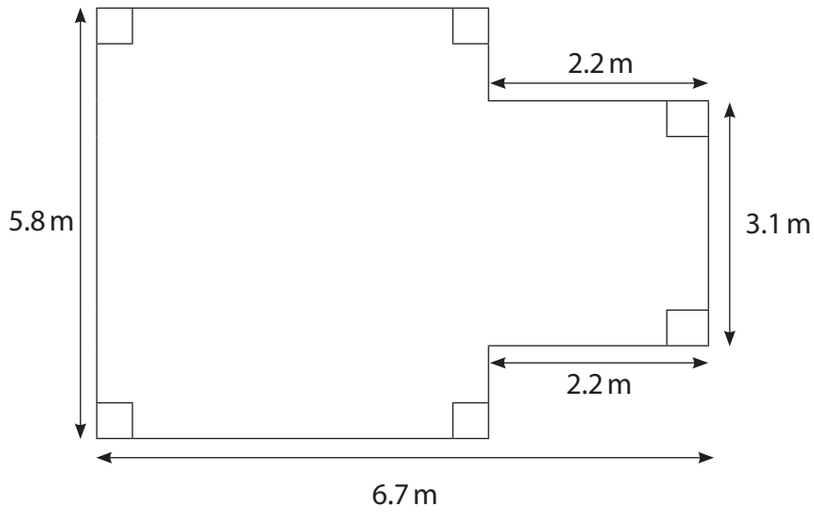


Diagram **not** accurately drawn

Cora allows 1m^2 of floor space for 2 people.

She thinks there is enough floor space in the room for 60 people.

(b) Is Cora correct?
Show why you think this. **(5)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate missing dimensions and split L-shaped floor into 2 rectangles *[M1 lengths]*

- **Total width = 6.7 m, Indent width = 2.2 m ⇒ Left width = 6.7 − 2.2 = 4.5 m**
- **Total height = 5.8 m, Right height = 3.1 m ⇒ Top indent height = 5.8 − 3.1 = 2.**

Step 2: Calculate total floor area (Vertical split into Rect A & Rect B) *[M1 area]*

- **Rectangle A (left section): 4.5 m × 5.8 m = 26.10 m²**
- **Rectangle B (right section): 2.2 m × 3.1 m = 6.82 m²**

Total Floor Area = 26.10 + 6.82 = 32.92 m²

Step 3: Calculate capacity (1 m² allows 2 people) *[M1 capacity]*

Total people allowed = 32.92 m² × 2 people/m² = 65.84 people

65.84 > 60 (room holds 65 people, exceeding target of 60)

Conclusion: Yes, Cora is correct (65.84 people ≥ 60 people). *[A1 decision]*

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Q7(b) Continuation: Alternative Floor Area Calculations & Capacity Check

Alternative Method 1: Subtraction from bounding rectangle

- Outer bounding rectangle: 6.7 m × 5.8 m = 38.86 m²
- Top-right cut-out corner: 2.2 m × (5.8 − 3.1) = 2.2 m × 2.7 m = 5.94 m²
- Floor Area = 38.86 − 5.94 = 32.92 m² (exactly matches)

Alternative Method 2: Horizontal split

- Bottom rectangle: 6.7 m × 3.1 m = 20.77 m²
- Top rectangle: (6.7 − 2.2) × (5.8 − 3.1) = 4.5 m × 2.7 m = 12.15 m²
- Floor Area = 20.77 + 12.15 = 32.92 m² (verified)

Required Area Method Check:

- Area needed for 60 people = 60 ÷ 2 people/m² = 30.00 m²
- Available room area = 32.92 m²

32.92 m² > 30.00 m² (surplus of 2.92 m² of floor space).

Final Answer:

Conclusion: Yes, Cora is correct.

Room Area = 32.92 m²; Max Capacity = 65 people (exceeds 60 people).

(Total for Question 7 is 9 marks)

8 Cora asked 720 people for their opinion of the catering at the venue.
Her results are shown in the table below.

opinion	excellent	good	poor
number of people	380	300	40

She wants to show this information in a pie chart for a report.

Cora starts to draw the pie chart.
She draws the largest sector.

Complete the pie chart below for Cora.

(3)

Use the box below to show clearly how you get your answer.

Step 1: Calculate angle per person in the pie chart (360° ÷ 720 people) *[M1 angle factor]*

Total people = 720; Total pie chart angle = 360°

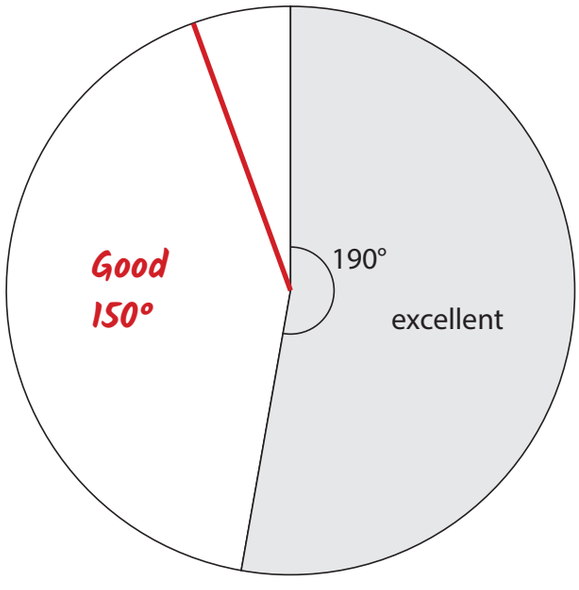
Angle per person = 360° ÷ 720 = 0.5° per person

Step 2: Calculate sector angle for each catering opinion *[M1 angles]*

- **Excellent (given): 380 × 0.5° = 190° (already drawn)**
- **Good: 300 people × 0.5° = 150°**
- **Poor: 40 people × 0.5° = 20° (Sum: 190° + 150° + 20° = 360°)**

Poor (20°)

[A1 complete chart]



(Total for Question 8 is 3 marks)

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9 Cora organises a wedding reception at the venue.
There will be 178 guests at the reception.

Cora knows that the ratio of waiters to guests should be 2 : 45

She thinks that she will need 8 waiters for the wedding reception.

Will 8 waiters be enough for 178 guests?
Show a check of your working. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Set up the ratio of waiters to guests (Waiters : Guests = 2 : 45) *[M1 ratio]*

Ratio = 2 waiters for every 45 guests (or 45 ÷ 2 = 22.5 guests per waiter)

Step 2: Calculate number of guests 8 waiters can serve (Method 1: Multiples) *[M1 calculation]*

8 waiters = 8 ÷ 2 = 4 pairs of waiters

Capacity for 8 waiters = 4 × 45 guests = 180 guests

Step 3: Compare guest capacity with 178 actual guests

180 ≥ 178 (8 waiters can serve up to 180 guests, which covers 178 guests)

Alternative exact decimal calculation (Method 2):

Waiters needed = (178 ÷ 45) × 2 = 3.9555... × 2 = 7.91 waiters

7.91 ≤ 8 (7.91 waiters needed is less than the 8 waiters available)

Conclusion & Final Decision:

Conclusion: Yes, 8 waiters will be enough (can serve 180 guests ≥ 178). *[A1 decision]*

Use the box below to show your check.

☑

Reverse calculation check: *[B1 check]*

180 ÷ 45 × 2 = 4 × 2 = 8 waiters (matches exactly)

Alternative check: 178 ÷ 8 = 22.25 guests/waiter; 22.25 < 22.5 max (verified)

(Total for Question 9 is 4 marks)

(TOTAL FOR PAPER IS 48 MARKS)

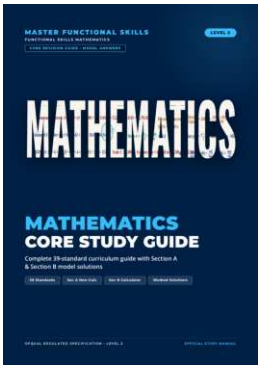
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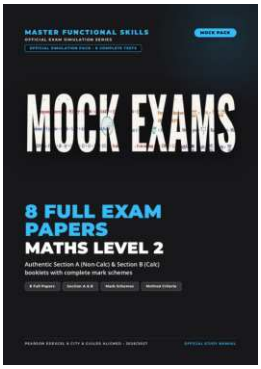


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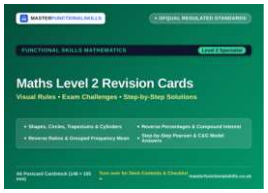


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